

ANAKHA S

Software Engineer — Python / Django · React / TypeScript · Applied AI & LLM Engineering

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SUMMARY

Software Engineer with a B.Tech in Computer Science and a Python full-stack internship, focused on backend systems, REST APIs, and applied AI engineering. Built a multi-tenant incident-response platform (Django REST Framework, React/TypeScript, PostgreSQL, Redis, Celery) with a provider-agnostic LLM pipeline, a custom semantic retrieval layer, role-based access control across four user roles, and 300+ automated tests running through a 3-pipeline CI setup. Comfortable across the stack — from Django ORM and asynchronous task design to React state management and LLM prompt/response validation.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL

Backend: Django, Django REST Framework, Celery, JWT Authentication (SimpleJWT), RESTful API design

Applied AI / LLM: OpenAI API, Anthropic API, Retrieval-Augmented Generation (RAG), embeddings, prompt engineering, semantic search

Frontend: React.js, React Router, TanStack Query, Zustand, Tailwind CSS, Vite

Databases: PostgreSQL, pgvector, Redis, MySQL

Tools & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), AWS (EC2, S3), Git, Pytest, Vitest, ESLint, Ruff, mypy

EXPERIENCE

Python Full Stack Developer Intern — Techolas Technologies | Kochi, Kerala | Jan 2025 – Nov 2025

- Built and maintained full-stack features across a Django/MySQL backend and a React + JavaScript frontend, shipping changes from API endpoint through UI.
- Optimized slow database queries and introduced view-level caching to cut redundant database calls on frequently accessed endpoints.
- Debugged production and staging defects and supported the release process for a live Django web application, from local development through deployment.

PROJECTS

AI Incident Response Platform — Personal Project

GitHub: <https://github.com/anakhas771/AI-Incident-Response-Platform>

- Built a multi-tenant, full-stack incident-management platform (React 18 + TypeScript frontend, Django REST Framework backend) spanning 7 backend modules and 33 REST endpoints, with organization-level data isolation enforced on every query.
- Designed a 4-role RBAC model (Admin, Analyst, Responder, Viewer) enforced through custom DRF permission classes that restrict AI-triggered actions to authorized roles while keeping read access open to the organization.
- Built a provider-agnostic LLM client (OpenAI, Anthropic, and a deterministic mock fallback) with exponential-backoff retries and structured JSON-schema validation, used to generate incident summaries, root-cause analysis, and remediation recommendations.
- Moved AI incident analysis off the request/response cycle into 13 Celery tasks with dedicated queue routing, exponential-backoff retries, and idempotent status tracking (pending → processing → completed/failed), backed by Redis.
- Implemented a custom semantic retrieval layer — document chunking, pluggable embedding providers, and cosine-similarity ranking with per-organization filtering — using pgvector in production with a JSON fallback for tests.
- Covered the codebase with 305+ automated tests across 54 modules, enforced through 3 GitHub Actions pipelines (backend, frontend, Docker); containerized the stack as a 6-service Docker Compose environment.

KnowledgeOS — Personal Project (in active development)

GitHub: <https://github.com/anakhas771/KnowledgeOS>

- Designed a multi-tenant Django REST Framework backend foundation for a planned enterprise knowledge platform, with organization-scoped data isolation and a custom 5-tier role model (Admin, Manager, Developer, Employee, Guest).
- Implemented JWT-based authentication (registration, login, token refresh) with custom token claims for role and organization, backed by a service layer that atomically creates an organization and its first user.
- Authored system, database, RAG-pipeline, and security architecture docs to plan the platform's retrieval-augmented assistant, and containerized PostgreSQL, Redis, and the API with Docker Compose, including a DB/cache health-check endpoint.

EDUCATION

B.Tech in Computer Science and Engineering — Indira Gandhi Institute of Engineering and Technology | Ernakulam, Kerala | 2021 – 2025

CERTIFICATIONS

Advanced Diploma in Artificial Intelligence & Machine Learning — Skill Development Council, Canada (2025)